

The effect of customers' AI-chatbot interactions on purchasing decisions: Dual mediating roles of customer digital intimacy and psychological ownership applied to Saudi SME customers

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ABSTRACT

This study aims to understand the way that AI chatbot interaction affects customers' purchasing decisions. This is achieved through testing the dual mediating effects of customer digital intimacy and psychological ownership on this relationship. The study utilized a quasi-experimental design to test the causal relationship. The data was collected from a sample of 360 customers of small and medium-sized companies in the Kingdom of Saudi Arabia. The study's model showed fully dual mediation effects of digital intimacy and psychological ownership on the relationship between AI chatbot interaction and purchasing decisions. AI chatbots show a significant positive effect on digital intimacy and a significant negative impact on psychological ownership. This study fills a gap in the literature by studying the dual mediating roles of digital intimacy and psychological ownership. It contributes to existing knowledge by looking at purchasing decisions as a psychological behavior that links customers to the product and the interaction media used.

1. Introduction

Encouraging the purchase of products and services is the ultimate goal of businesses. All business endeavors serve the interests of customers' purchasing decisions (Cao et al., 2025). Traditionally, businesses have interacted with their customers through sales and customer service teams. Sales teams are trained to influence customers' purchasing decisions. Humans can learn and exhibit innovative behavior to accommodate customers' desires and needs. Recently, communication and interactions with customers have been carried out with the use of automated tools such as programmed and AI-based chatbots (Jadhav et al., 2022). The efficiency and effectiveness of these tools in influencing customer purchasing decisions are the subject of ongoing debate in the literature (Sun et al., 2023).

One innovative tool for influencing purchasing decisions is customer relationship management technologies such as AI-based chatbots (Jadhav et al., 2022). It is assumed that new, low-cost technologies such as AI chatbots (Marcinekova et al., 2025) can positively affect customers' purchasing decisions, yet this is not the case. These technologies can sometimes lead to adverse outcomes (Hasal et al., 2021). This can be attributed to human anthropomorphism (Ku, 2024). One caveat to AI

chatbots' success in influencing customers' purchasing decisions is their ability to create a warm, comfortable feeling in customers and to make customers feel that the product is theirs (namely, the interaction side and the product side). Customers need to be interacted with based on their feelings and needs. The main interest of this study is to understand this research problem.

Small and medium-sized companies have fewer financial resources than larger companies. Thus, they are forced to invest these resources more efficiently. From this point of view, small and medium-sized businesses are motivated to invest in new technologies such as AI chatbots to help them in cost reduction (Marcinekova et al., 2025). Yet to harvest the best of these technologies, they should be built on customers' comfort, interest, and trust (Wang et al., 2025). In addition, they should be designed to be good product promoters. From this perspective, we need to understand whether new technologies such as AI chatbots can create digital intimacy and psychological ownership for customers.

This study aims to examine the effect of AI chatbot interactions on customers' purchasing decisions. This main effect is proposed to be affected by the nature of the interaction between humans and technology. This study presents this interaction by integrating two mediating variables: digital intimacy and psychological ownership. It contributes

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to existing knowledge by shedding light on the interaction between the AI chatbots and humans. It also demonstrates existing AI chatbot technology's effectiveness and its ability to foster digital intimacy and psychological ownership among customers.

This study is organized into five sections. The first section introduces the study. The second section is devoted to analyzing the literature review. The methodology of this study is explained in the third section. The fourth section presents the findings and the data analysis, while the fifth section contains the discussion and the conclusion of the study.

2. Literature review

2.1. Purchasing decisions

A purchasing decision is when a consumer chooses to buy a product or service and executes on the decision. Previous studies have focused on purchasing intention, which can be affected by the psychological behavior of the customers, and thus, group membership can influence this decision (Zhang et al., 2025). Customers also place weight on the perceived benefits of the product or service (Abdul Latip et al., 2025). More importantly, they also look for trust in product assurance, which shapes their purchasing intention (Abdul Latip et al., 2025). Researchers have also noted that intention to purchase electronic products is influenced by electronic word of mouth (Abdul Latip et al., 2025) and knowledge of the product (Abdul Latip et al., 2025; Liu et al., 2025). Moreover, other customers' reviews and the attractiveness of the product title play a role in this domain (Bai et al., 2025; Xu et al., 2025).

Although only a few studies have investigated actual purchasing decisions, they identified several noticeable factors. The effect of customers' social life has been acknowledged in the literature, such as the choices of the customers' children (Duus et al., 2025), the availability of the product (Duus et al., 2025), the distribution strategy and warehousing strategies (Pratiwi et al., 2025), pricing strategy (Pratiwi et al., 2025), rewards (Pratiwi et al., 2025), and heuristics, information, and labels (Jacobs et al., 2025; Pienwisetkaew et al., 2025). Others have discussed the lack of information regarding the delay of purchasing decisions (Benhissi & Hamouda, 2025).

Interestingly, people tend to spend extravagantly on purchasing products that give them emotional support and happiness (Liu, Chang, et al., 2024). Thus, treating the purchasing decision as solely performed to fulfill a customer's needs is not correct. Purchasing decisions are psychological activities and should be studied from this perspective.

2.2. Using technology

The effect of technology use on purchasing decisions or the intention to purchase has also been investigated. Previous studies found that using technologies and automation help customers make purchasing decisions for sensitive products that require greater privacy (Cardinali et al., 2025). In addition, using more sophisticated AI technologies increases empathy, helping reduce the length of purchasing decision postponement (Li & Kang, 2025). Moreover, using a more familiar technology interface can help shape purchasing decisions (Palaniappan et al., 2025). Using technologies such as QR codes and blockchain helped customers obtain more information about the products, build trust, and satisfy their information hunger, which, in turn, encouraged them to buy (Adamashvili et al., 2024). In summary, customers are ready for technology and like when it is involved in their purchasing decisions.

Purchasing decision is clearly a complicated behavior that should be properly understood. Although a customer's purchasing decision is about satisfying human needs, it is, without doubt, also an emotional and psychological behavior (Ngo et al., 2026). Using technology such as AI chatbots should take this into account. Human emotions should be considered when introducing AI chatbots as a low-cost marketing and interaction tool in order to secure more successful applications for these innovative technologies.

2.3. AI chatbots

AI chatbot is a type of artificial intelligence software intended to chat and communicate with people (Albayrak et al., 2018). These bots are used in businesses to offer better customer service. They are designed to be faster and more informative than human customer service and to help customers make purchasing decisions (Albayrak et al., 2018). Businesses can deploy these bots on their websites or integrate them into popular platforms such as Facebook (Krishnan et al., 2022). Thus, they are considered flexible tools that can be used across a wide range of businesses and platforms.

The literature discusses several important applications of AI chatbots. In the education context, chatbots were found to help create supportive learning environments (Mohamed et al., 2025) and to affect students' engagement, interaction, and academic performance (Ait Baha et al., 2024). These tools often can't be avoided and should be used in education, provided that a solid ethical framework is created and followed (Kooli, 2023). Chatting with AI chatbots improves language learning among students outside the classroom and reduces fear of judgment (Zhang, 2026). AI chatbots can also help teach applied science (Rücker & Becker-Genschow, 2025) and improve students' belief in their ability to succeed more than other AI tools (Naamati-Schneider & Alt, 2025).

In education, AI chatbots can be a double-edged sword. Although they can improve students' performance, they can also worsen some learning aspects (Gu & Liu, 2025). This indicates that learners misunderstood the nature of communication with AI chatbots. A clear indication is a previous study suggesting that students do not use AI chatbots mainly because of their usefulness, but as a response to social norms (Ballesteros et al., 2025). This is a strong indication of the psychological behavior that could affect the relationship between AI chatbots and students.

In the healthcare services context, AI chatbots have been found to offer easy access to patients and to be supportive, yet there is no clear evidence of their effectiveness. What is interesting here is that previous studies thought that AI chatbots would be effective; they needed to understand human speech (Milne-Ives et al., 2020). AI chatbots can be a source of information for patients (Merç et al., 2026) and can help patients adhere to their medication plans and educate them more about their conditions, yet risks include over-reliance, weakening patient relations with a human therapist, and the inability to imitate human empathy (Denecke et al., 2021).

In the business context, AI chatbots have been used to improve business growth through customer engagement (Krishnan et al., 2022). They can also facilitate communication with employees (Pillai et al., 2024). However, employees have contradictory experiences; some think AI chatbots know what they need and are quicker to answer their queries, imitate human experience, and show smartness, while others are unsure about data safety, doubt their understanding of these bots, or are not familiar with the technology (Pillai et al., 2024).

Human interaction with AI chatbots should be understood through emotions and psychological behaviors. For example, AI chatbots can be confusing and seen as lacking empathy for customers' perspectives (Marcinekova et al., 2025). Moreover, chatting with AI chatbots in a political voting context showed that voters felt a personal connection with the AI chatbots, yet they lacked trust in them (Lee & Yuan, 2025). More concerningly, chatting with AI chatbots was found to cause psychological problems (Richet, 2025). On the other hand, they can reduce anxiety (Fattouh, 2025). From a business perspective, having the AI chatbot talk more to customers and give them more information seems to be a successful strategy (Yu et al., 2025).

2.4. Digital customer intimacy

Digital customer intimacy is the ability of a digital platform to create a feeling of warmth and belonging among people who interact with it

(Liu, Chan, & Chimhundu, 2024). Digital customer intimacy is a distinct concept that not only aims at having a positive attitude toward the product, but it is also about improving the interaction with new technology. It is also about the creation of a humanized experience with new technologies (Liu, Chan, & Chimhundu, 2024). Digital customer intimacy is a much wider concept than customer trust. Customer trust is just a bridge between humans and the digital platform, while digital customer intimacy is an emotional framework that harmonizes this relationship. Digital customer intimacy should not be confused with customer rapport. Customer rapport is just the initial step of the relationship between customers and digital platforms, while digital intimacy is the motivation for continuing this relationship for a long time period. Moreover, digital intimacy presents much deeper behavior compared to customer engagement. Customer engagement is the motivation for more frequent or repeated interaction, while digital intimacy is about how deeply they feel that the other side can understand their feelings and emotions.

Digital customer intimacy has become an important concept as businesses increasingly rely on digital platforms and e-commerce to sell their products and services. As discussed earlier, a purchasing decision is an emotional one, and thus, digital customer intimacy can be a significant influencing factor. This can be discerned intuitively from the psychological effect of online ads on buying decisions (Xu et al., 2024). Even for rational and self-confident customers, aggressive or annoying ads can affect or reverse their purchasing decisions. In addition, simply listing products in an app doesn't persuade customers to buy them; the app should be designed to be informative and supportive, helping customers alter their purchasing decisions (Lu et al., 2025).

AI intimacy depends on a complex relationship shaped by small changes in moral interaction and in the nature of humans, such as the immersion tendency (Bao et al., 2026). Humans can experience real and genuine intimacy in their interactions with machines and the internet (Smith & Snider, 2019), and this type of AI intimacy creates trust, satisfaction, and emotional dependence (Bao et al., 2026). Additionally, customers interact better with AI agents when they look attractive, have solid expertise about the product or the service, and have congruence (Zhang & Zhu, 2025). This shows the customers' need to create digital intimacy with these platforms.

Previous studies discuss the importance of AI intimacy in the use of AI chatbots as commercial tools. AI intimacy can be used to improve service competencies (Ku, 2025). Indirectly, previous research showed the importance of customer digital intimacy by demonstrating that digital aesthetics and playfulness affect purchase intention through providing customers with a pleasurable experience (Rahman et al., 2026). Interaction with AI robots possessing perceived warmth can affect customer satisfaction through digital marketing innovation (Ku, 2024). Marketing apps should create a feeling of intimacy to create brand citizens instead of aiming only to achieve more sales (Garrouch & Ghali, 2023), and therefore using digital interaction tools such as an AI avatar while understanding the need for human intimacy can create loyal customers (Shah et al., 2025).

2.5. Psychological ownership

Psychological ownership is the feeling of owning a product or a service without officially owning it (Scarpi, 2024). Psychological ownership is developed through three main antecedents. First is the feeling of mineness because people feel that they can touch it or decide what happens to it. The second antecedent is the feeling of ownership because people invest their time or effort in that thing. The third antecedent is the feeling of having shared history with it (Morewedge, 2021). It is a powerful and defensive behavior that can be linked to feelings of self-protection (Hingston & Whelan, 2024) and that can be developed before the purchasing decision occurs (Mahimkar, 2026). Psychologically, people exaggerate the value of the products they own. Thus, when customers develop psychological ownership but do not buy the product,

their feelings shift from considering it a gain to a loss. Customers with psychological ownership treat a product as an extension of themselves (Nguyen et al., 2025). Psychological ownership increases the desirability of a product or service, the willingness to pay more for it, and the customer's sense of responsibility toward it (Alshammari & Al-Mamary, 2026).

There is an urgent need to test the ability of AI-based technology to foster a sense of psychological ownership of its products when interacting with customers. The interaction between customers and human customer service agents can effectively increase psychological ownership (Jiang et al., 2026). However, the literature questions chatbots' ability to create psychological ownership because they are pre-programmed and thus lack autonomy or learning abilities (Scarpi, 2024). This skepticism does not apply to AI chatbots, as they can learn and create autonomous responses.

2.6. Research gap

Research on AI chatbots in sales and marketing is still in its early stages. We need more research to better understand this issue. The most critical factor here is the interactions between customers and AI chatbots (Kumar et al., 2025). Previous studies have focused on several factors that can affect purchasing decisions, but they have not investigated the way or mechanism by which these factors influence the purchasing decision. This study fills this gap by highlighting the importance of the interactions between humans and technology in shaping purchasing decisions. This study proposed two main mediating factors, digital customer intimacy and psychological ownership, as contributing factors through which AI chatbots can affect purchasing decisions.

2.7. Conceptual framework

As we explained the necessity of understanding the research problem and addressing the literature gap related to psychological aspects of customers' purchasing decisions due to the interaction between the customer and the AI chatbots, this study proposed the following conceptual framework. The conceptual framework shows two effects of AI chatbot interaction on purchasing decisions. The first one is the direct effect of the AI chatbot interaction on customers' purchasing decision. The second effect is the indirect effect of AI chatbot interaction on purchasing decisions mediated by digital intimacy and psychological ownership (Fig. 1).

3. Research methods

3.1. Research design

The current study implemented a quasi-experimental, quantitative, exploratory design to test the main effect of AI chatbot interaction on purchasing decisions and the dual mediating effects of digital customer intimacy and psychological ownership. This quantitative design utilized propensity score matching (PSM) to control for systematic differences between groups in a set of confounders (Caliendo & Kopeinig, 2008). PSM matches cases in the treatment group with those in the control group based on each case's propensity scores. Thus, PSM strengthens the causal argument in our observational study by reducing selection and confounding bias (Lu et al., 2018). We used the nonparametric pre-processing for parametric causal inference (Salditt & Nestler, 2023). Propensity score matching was implemented using a ratio of 1:1. The main method was implemented using the nearest neighborhood method, utilizing a generalized linear model to fit the logistic regression technique. A set of six control variables was used in performing the PSM. PSM design was applied based on natural assignments to treatment and control groups, without any manipulation by the researchers. The customers were selected without any clue about their status. The assignment to the treatment and control groups was based on customer

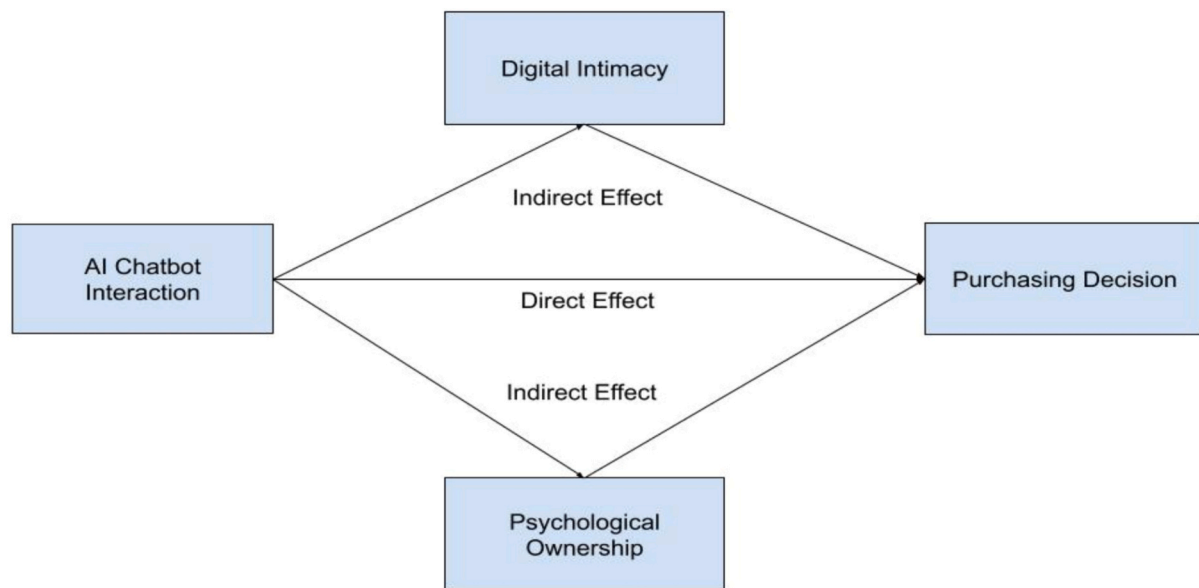


Fig. 1. Conceptual framework.

feedback. The balance of the data in the PSM was tested using the standardized mean difference between the control and treatment groups. The standardized mean difference has a cutoff of 0.1. Any score higher than 0.1 is considered imbalanced (Zhang et al., 2019).

The analytical technique used is partial least squares structural equation modeling (PLS-SEM), which is based on variance analysis and is the best choice when exploring and testing causal relationships, as it maximizes explained variance for predictive purposes (Hair, 2021). Thus, PLS-SEM was found to be ideal to explore the relationships between AI chatbot interactions, digital intimacy, psychological ownership, and purchasing decisions. PLS-SEM is found to be the best analytical tool in our case because of three advantages. The ability of the latest PLS-SEM software to handle binary variables without even the need to standardize them (Becker et al., 2022). Thus, we used SmartPLS 4 for this reason (Christian et al., 2024). PLS-SEM is a much more credible option to analyze complex models with moderating or mediating effects (Hair, 2021). Like any other structural equation modeling technique, PLS-SEM accounts for measurement error (Hair, 2021). Counting for measurement errors is essential for the psychometric scales used in this study, such as digital intimacy and psychological ownership. There is no way to account for measurement error in other competing techniques, such as linear or logistic regression.

3.2. Variables measurement

3.2.1. Digital customer intimacy

Digital customer intimacy was measured using a 12-item scale adopted from (Sternberg, 1997). This scale measurement is ideally suited to the purpose of this study (i.e., measuring the intimacy and comfortable relationships that may develop between customers and AI chatbots). This scale measures intimacy in the sense of creating a supportive partnership rather than romantic feelings. Sternberg's triangle love scale contains three major components: intimacy, passion, and commitment. The three components are meant to construct a love scale. Our interest was in measuring intimacy rather than love. The intimacy component contains 12 flexible items with spaces to fill in whatever you want to measure about intimacy. The scale is very simple when it comes to wording. Thus, we didn't find any culturally or contextually specific words that would create confusion for local Saudi respondents. Thus, we adopt the scale as it is. We just filled in the required empty spaces. For example, Item number 11 is "I feel emotionally close to ____", was

adopted as "I feel emotionally close to the sales AI chatbot when using it". Because this scale has no reverse items, we reversed item number 5, "I received considerable emotional support from the sales AI chatbot of this company". In addition, for item 3, "I strongly desire to promote the well-being of the sales AI chatbot," we kept it as is, but added an example to clarify the meaning of "well-being."

3.2.2. Psychological ownership

Psychological ownership was measured using a scale adopted from (Van Dyne & Pierce, 2004). This scale is widely used to measure psychological ownership in different contexts. The scale consisted of seven items, one of which had a reverse meaning.

3.2.3. AI chatbot interactions

The independent variable was measured as a binary (0/1) variable. The zero value indicates interaction with a human, while a value of 1 indicates interaction with an AI chatbot. Customers initiate an interaction when they intend to purchase products or services. The chatbot interaction is mainly used to ask for information about the product or service, such as features, pictures, ingredients, quality, comparisons with other products, the suitability of the product for the customer, etc. Both human and AI chatbot interactions are aimed at stimulating purchasing decisions. There is no specific, fixed timing between the interaction and the purchasing decision. Sometimes a purchasing decision comes directly after interaction, especially when it is done online. In some cases, it can take a while, and in others, it may require more than one session of interaction. In some other cases, it never happens.

3.2.4. Purchasing decisions

The independent variable was measured as a binary (0/1) variable. The zero value indicates the customer did not purchase any products or services from the company, while a value of 1 means the customer purchased products or services as a result of an interaction with an AI chatbot or a human. The selection of the binary scale was based on the objective of the study, which focuses on the actual action of the purchasing decision. Previous studies didn't focus on the actual purchase and instead focused on other status of the purchasing decisions, such as the intention to purchase. The actual purchase is best captured by whether they bought a product or not. In addition, the binary scale has no less power to produce the same conclusion as the other scale formats. Moreover, the binary scales are much faster to complete and produce

more reliable answers (Dolnicar & Grün, 2007).

3.3. Control variables

The first control variable was age. Age is a key factor in this study, as different generations have different familiarity with and usage of new technologies. Younger generations use new AI-based technology more frequently than older generations. The second control variable was product involvement (Bauer et al., 2006). Customers with varying product involvement can behave differently when purchasing products and services. The third control variable was income. Different levels of income can affect the purchasing decision. Gender, education, and AI literacy were also included (Grassini, 2024).

3.4. Sampling and data collection procedure

The study population consisted of residents of the Kingdom of Saudi Arabia who are able to purchase products and services from small and medium-sized companies, which are defined as any company with 249 employees or fewer and with an income of less than 50 million US dollars (200 million Saudi Riyals). A questionnaire was developed based on well-defined scales and distributed electronically. The link was set to 15 surveyors, who were responsible for distributing it to the targeted respondents. These surveyors were located in 6 main areas of Saudi Arabia and were able to reach respondents in other regions easily through online tools. The surveyors contacted the customers of SMEs through social media platforms or face-to-face at some of the companies' retail stores and showrooms. On social media, customers were selected based on their interactions, likes, feedback, questions, or criticism of the products or services. In the stores, customers were selected based on a mall intercept sampling technique: the surveyors asked customers about their desire and ability to fill in the questionnaire, and if the customer accepted the invitation, a link to the digital questionnaire would be sent to them. At their convenience, the customer filled out the questionnaire. No pressure was put on customers to complete the questionnaire. The screening question technique was used to establish the causal effect of the interaction on the purchasing decision. The procedure is established during the data collection stage by asking customers whether their purchasing decision was influenced by their interaction with the company. After they confirmed that, the customers were asked about the type of interaction, whether it was with the AI chatbot or with the sales teams.

3.5. Pilot study

A pilot study was conducted on a group of 38 customers, 20 females and 18 males. The pilot study tested the applicability of the data collection plan. These customers are known personally to the authors of the study. The respondents were chosen and fulfilled the study's selection criteria. Seventeen respondents completed the questionnaire in less than 5 min, nine took less than 8 min, and the remaining twelve respondents took around 10 min. There were no complaints about the clarity of the questions, but 23 of the respondents felt annoyed by the questions. When we discussed these problems, we discovered that the discomfort stemmed from the reverse questions and the marker variable section, which is to be expected. Although this option was not available during the pilot study, 36 respondents preferred to receive the questionnaire link and complete it later at their convenience. This was very helpful information for conducting the main study. All of the respondents expressed that their most comfortable time to complete the questionnaire was between 5 PM and 9 PM.

3.6. Bias checks

Common method variance was assessed using the marker variable method. The best marker identified is the attitude toward the color blue

(Miller & Simmering, 2023). The marker variable was measured using the shorter version of the Miller and Simmering (2023) scale. This scale consists of three items: "I like the color blue", "The color blue is one of my favorite colors", "I have a positive attitude toward the color blue". This scale was measured using a 5-point Likert scale. The test is performed by regressing the marker variable on all the variables presented in the model. This step should not significantly change the effect of the independent variables and the mediating variable on the dependent variable. The test showed no major problem with common method bias (see the data analysis section).

Social desirability bias was controlled by making all respondents' identities anonymous. Acquiescence bias was controlled by using reverse wording items. The loading of these items was acceptable, as shown in the measurement model tests. Non-response bias was tested by comparing early and late responses (Armstrong & Overton, 1977). Non-response bias is measured by comparing the characteristics of the respondents and the non-respondents. Surveyors gave customers a free choice to fill in the questionnaire or not. In the case of refusal or unwillingness to fill in the questionnaire, the surveyors asked customers to share only their demographic data. Most of the non-respondents refused even to share their demographics, except for 51 customers. This group of non-respondents was used to test the non-response bias by comparing their characteristics with those of the respondents. Among the non-respondents, there were 34 females and 17 males; 41 held bachelor's degrees, 9 held master's degrees, and only 1 respondent held a doctorate degree. Of the 51 respondents, 36 were aged 18–35, and 15 were aged 36–55. In terms of income, 28 respondents had an income between 5000 and 10,000 SR, 22 had an income between 10,000 and 20,000 SR, and only 1 had an income of more than 20,000 SR. These results show that there was no systematic non-response bias. Endogeneity bias was eliminated by using a quasi-experimental design with propensity score matching. The mechanism of propensity score matching controls for confounding effects.

3.7. PLS-SEM model fitness checks

PLS-SEM models need to be tested for their ability to fit the data. The tests include the measurement and structural models. The measurement model is called the outer model, and it consists of the relationships between the latent variables and their measured items. The loadings of the measurement models should show a score of 0.7 or more. Testing the internal consistency was measured using two techniques, Cronbach's alpha and composite reliability tests. Cronbach's alpha and composite reliability test scores should exceed the threshold of 0.7; if not, it is considered unacceptable (Hair, 2021). Discriminant validity was conducted to confirm that each group of items is able to measure a distinct latent variable. For this test, the HTMT ratio is used (Hair, 2021). HTMT scores should show scores that are less than 0.85 to be accepted. Another test, called convergent validity, tests the relationship between latent variables and their measurement to confirm the fit between them. The convergent validity is tested using the average variance extracted (AVE), which should show more than 0.5.

3.8. Ethical consideration

This study did not experiment directly on humans or animals. The data collected in this study showed no adverse effects on the respondents that would create any harmful or uncomfortable conditions, results, or circumstances. When the data were collected, we observed a high level of anonymity, so no privacy issues arose. Respondents' contributions to this study were entirely voluntary, and no enforcement measures were used. We declared that all of the data collected for this study would not be used for any purpose other than research. Thus, the data collected in the survey did not harm the respondents in any foreseeable way. Ethical approval for conducting this study was obtained from the Research Ethics Committee (REC) at the University of Hail, Saudi Arabia, and

registered under the number 158-L-08-H.

4. Data analysis and findings

4.1. Sample descriptives

The sample was 50.4% female and 49.6% male. In terms of education, 56.9% of respondents held a bachelor's degree, 29.2% held a master's degree, 8.2% held a doctoral degree, and 5.7% were categorized as other levels of education. The age analysis showed that 47.4% of respondents were 18–35 years old, 43.1% were 36–55 years old, and 9.5% were aged 56 or older. The income analysis showed that 36.7% of respondents had a monthly income between 5000 and 10,000 Saudi Riyal, 37.9% between 10,000 and 20,000 SR, 12% less than 5000 SR, and 13.4% had a monthly income of more than 20,000 SR (exchange rate is fixed to 3.75SR/\$). As part of the purchasing decision, 49.6% respondents interacted with AI chatbots while 50.4% interacted with marketing/sales team members. In addition, 37.1% of respondents did not purchase the product/service, while 62.9% purchased the intended product or service as a result of the interaction with the companies. On the organizational level of the small and medium businesses, it was found that, 19% were in cosmetics and perfumes businesses, 17% were clothes and fashion businesses, 14% were electronics, computers, and mobile business, 11% were jewelry business, 10% were selling furniture and home appliances, 9% were food and restaurant business, 8% were pharmaceutical products and supplements business, 7% were selling fitness services, while 5% were in the education and training services. The majority of these businesses deployed their AI chatbots on the WhatsApp Business API and Instagram. These platforms are powered by a local platform named Gabster AI and Humanin chat, as these service providers specialize in Arabic models.

4.2. Propensity score matching analysis

Propensity score matching was applied to the available data to eliminate potential confounding effects on the relationship between AI chatbot interaction and the purchasing decision. As a result, 360 data points were satisfactorily matched (Table 2). The propensity scores matching shows that there were some potential unbalanced data related to three variables, gender, age, and AI literacy, with standardized mean differences of 0.20, 0.18, and 0.15, respectively. This resulted in an overall distance with a standard mean difference of 0.16. The propensity score matching eliminates this potential data imbalance and reduces the three variables' standardized means to 0.07, 0.09, and 0.08, respectively. The overall distance standardized mean difference is reduced to 0.10, as shown in Table 1.

4.3. Common method bias

Table 3 shows that adding the marker variables does not have a significant impact. The effect of chatbot interaction on digital intimacy changes slightly from 0.736 to 0.710 and is still significant. The effect of chatbot interaction on psychological ownership is still significant, while

Table 2

Final sample size after applying PSM.

	Control	Treated
All	190	187
Matched	180	180
Unmatched	10	7
Discarded	0	0

Table 3

Common method bias tests.

	With marker		Without marker	
	Original sample (O)	P values	Original sample (O)	P values
Chatbot interaction - > digital intimacy	0.714	0.000	0.7360	0.0000
Chatbot interaction - > psychological ownership	−0.655	0.000	−0.6240	0.0000
Chatbot interaction - > purchasing decision	0.045	0.400	0.0510	0.3340
Digital intimacy - > purchasing decision	0.197	0.000	0.2000	0.0000
M1 - > chatbot interaction	−0.049	0.062		
M2 - > digital intimacy	−0.113	0.024		
M3 - > psychological ownership	−0.157	0.001		
M4 - > purchasing decision	−0.029	0.200		
Psychological ownership - > purchasing decision	0.080	0.002	0.0840	0.0010

changing slightly from −0.624 to −0.655. The effect of chatbot interaction on purchasing decision is still insignificant, while changing slightly from 0.051 to 0.045. Digital intimacy had a significant impact on purchasing decisions before and after adding the marker variable and only changed from 0.2 to 0.197.

4.4. Measurement model analysis

Checking the measurement showed a satisfactory fit of the latent variables and their measurement items. The loading of the digital literature construct ranged from 0.875 to 0.894. The loadings of psychological ownership also showed an excellent measurement model, varying between 0.738 and 0.884. All of these loadings exceed the threshold of 0.705 (Fig. 2).

4.5. Reliability and validity analysis

The reliability and validity analysis also showed satisfactory scores. Starting from the internal consistency measure of Cronbach's alpha, digital intimacy scored 0.90, while psychological ownership scored 0.89. The composite reliability showed a similar pattern, with a score of 0.906 for digital intimacy and 0.902 for psychological ownership. All of these scores exceed the required threshold of 0.7. The average variance extracted showed good fit, as digital intimacy scored 0.719 and psychological ownership scored 0.691. Both scores exceed the threshold of

Table 1

Propensity score matching scores.

	Before			After		
	Means treated	Means controlled	Std. mean diff.	Means treated	Means controlled	Std. mean diff.
Distance	0.5632	0.4624	0.1632	0.5431	0.484	0.1058
Gender	0.6385	0.3827	0.2017	0.5889	0.4333	0.0722
Education	2.3951	2.4157	−0.0388	2.3778	2.4111	−0.0575
Age	1.4512	1.8112	−0.1815	1.5111	1.7056	−0.0961
Income	2.5408	2.5476	−0.0168	2.5167	2.5222	−0.0153
Product involvement	2.7537	2.8463	−0.0479	2.8019	2.8796	−0.0327
AI literacy	3.4804	3.1277	0.1587	3.4213	3.2176	0.0863

Table 4

Measurement model loadings and reliability and validity tests.

	Loadings	Cronbach's alpha	Composite reliability	Average variance extracted (AVE)	VIF
DI1 < - digital intimacy	0.875	0.903	0.906	0.719	4.253
DI10 < - digital intimacy	0.890				4.721
DI11 < - digital intimacy	0.878				4.866
DI12 < - digital intimacy	0.885				4.858
DI2 < - digital intimacy	0.882				4.517
DI3 < - digital intimacy	0.846				3.515
DI4 < - digital intimacy	0.873				4.020
DI5 < - digital intimacy	0.879				4.387
DI6 < - digital intimacy	0.863				3.752
DI7 < - digital intimacy	0.877	0.891	0.902	0.691	4.178
DI8 < - digital intimacy	0.894				4.636
DI9 < - digital intimacy	0.880				4.179
PO1 < - psychological ownership	0.883				3.495
PO2 < - psychological ownership	0.879				4.022
PO3 < - psychological ownership	0.882				3.983
PO4 < - psychological ownership	0.875				3.957
PO5 < - psychological ownership	0.875				4.004
PO6 < - psychological ownership	0.884				3.520
PO7 < - psychological ownership	0.738				1.859
CI < - chatbot interaction	1				1.000
PD < - purchasing decision	1				1.000

Table 5

Heterotrait-monotrait ratio (HTMT) – Matrix.

	Chatbot interaction	Digital intimacy	Psychological ownership	Purchasing decision
Chatbot interaction				
Digital intimacy	0.371			
Psychological ownership	0.320	0.147		
Purchasing decision	0.151	0.461	0.223	

Table 6

Dependent variables R-squared scores.

	R-square	R-square adjusted
Digital intimacy	0.135	0.133
Psychological ownership	0.097	0.095
Purchasing decision	0.234	0.228

Table 7

Path coefficients of the structural model.

	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T statistics (O/STDEV)	P values
Chatbot interaction - > digital intimacy	0.736	0.740	0.094	7.840	0.000
Chatbot interaction - > psychological ownership	-0.624	-0.630	0.096	6.507	0.000
Chatbot interaction - > purchasing decision	0.051	0.053	0.053	0.965	0.334
Digital intimacy - > purchasing decision	0.200	0.200	0.025	8.144	0.000
Psychological ownership - > purchasing decision	0.084	0.085	0.026	3.285	0.001

Table 8

Specific indirect effects coefficients.

	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T statistics (O/STDEV)	P values
Chatbot interaction - > psychological ownership - > purchasing decision	-0.053	-0.054	0.020	2.638	0.008
Chatbot interaction - > digital intimacy - > purchasing decision	0.147	0.148	0.025	5.823	0.000

Table 9

Specific Indirect effect coefficients confidence interval.

	Original sample (O)	Sample mean (M)	2.5%	97.5%
Chatbot interaction - > psychological ownership - > purchasing decision	-0.053	-0.054	-0.098	-0.019
Chatbot interaction - > digital intimacy - > purchasing decision	0.147	0.148	0.100	0.199

Table 10

Total indirect effects' coefficient.

	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T statistics (O/STDEV)	P values
Chatbot interaction - > purchasing decision	0.094	0.094	0.037	2.580	0.010

0.5. The last test was the variance inflation factors, which showed results as expected. The items of the constructs should be highly correlated, yet not exactly the same. Digital intimacy VIFs varied between 4.0 and 4.85, while psychological ownership VIFs varied between 3.49 and 4.86. All

Table 11
Total effect coefficient confidence interval.

	Original sample (O)	Sample mean (M)	2.5%	97.5%
Chatbot interaction - > purchasing decision	0.094	0.094	0.020	0.164

of these scores are less than 5. See Table 4.

Discriminant validity is tested using the Heterotrait-Monotrait ratio (HTMT). The test confirmed the discriminant validity of the different constructs used in this study, as all of the scores are less than 0.85. The ratio between chatbot interaction and digital intimacy is 0.371. The ratio between chatbot interaction and psychological ownership is 0.32. The ratio between chatbot interaction and purchasing decision is 0.151. The ratio between digital intimacy and psychological ownership is 0.147. The ratio between digital intimacy and purchasing decisions is 0.461. Finally, the ratio between psychological ownership and purchasing decision is 0.223. Table 5 shows the results of the HTMT tests.

4.6. Structural model

A structural model was used to test the relationships between the main variables of this study. It tests the dual mediating effects of digital intimacy and psychological ownership on the main relationship between AI chatbot interaction and the purchasing decision. The first step was to test the total variance of the dependent variable explained by the independent variables. The model explains around 14% of the variation in the digital intimacy variable, around 10% of the variation in the psychological ownership variable, and 23% of the variation in the purchasing decision variable. The second step was to test the significance of the relationships among the variables of the model. A summary of the results is shown in Table 6.

Tests showed that AI chatbot interaction had a positive and significant impact on digital intimacy with a score of ($B = 0.736$, T Statistic = 7.84). Chatbot interaction had a negative and significant impact on psychological ownership with a score of ($B = -0.624$, T Statistic = 6.507). Digital intimacy had a positive and significant impact on purchasing decisions with a score of ($B = 0.2$, T Statistic = 8.14). Psychological ownership had a positive and significant impact on purchasing decisions with a score of ($B = 0.08$, T Statistic = 3.285). The direct

impact of chatbot interaction had an insignificant impact on purchasing decision with a score of ($B = 0.05$, T Statistic = 0.965). These results are shown in Table 7.

4.6.1. The indirect effects

The chatbot interaction shows a positive and significant impact on purchasing decision through digital intimacy with a score of ($B = 0.147$, T-Statistic = 5.823). The 95% confidence interval indicated that the true impact score would fall between 0.1 and 0.199. On the other side of the mediation effect, chatbot interaction shows a negative and significant impact on purchasing decisions through psychological ownership, with a score of ($B = -0.054$, T-statistic = 2.638). The 95% confidence interval indicated that the true impact score would fall between -0.098 and -0.019 (Tables 8 and 9).

4.6.2. Total effect

The chatbot interaction shows positive and significant total impact on purchasing decision through the direct impact, digital intimacy, and psychological ownership, with a score of ($B = 0.094$, T-Statistic = 2.58). The 95% confidence interval indicated that the true impact score would fall between 0.020 and 0.164 (Tables 10 and 11).

5. Discussion

This study investigated the psychological interaction between machines and humans with the specific objective of investigating the impact of interactions between customers and AI chatbots on purchasing decisions. This relationship was studied in the presence of two main mediating factors: digital intimacy and psychological ownership. Results showed that purchasing decisions are psychological decisions, and interactions with customers should be established with this principle as a foundation. Investments in technologies to support interactions with customers should prioritize customers' psychological behavior. These technologies should be built to improve intimacy with humans and create psychological ownership, as without these features, technology such as AI chatbots are not beneficial.

The findings of this study showed that interaction with AI chatbots in Saudi small and medium-sized companies was able to improve digital intimacy. Digital intimacy can help in creating a comfortable atmosphere between customers and AI chatbots. This shows that humans can develop warm feelings toward machines such as AI chatbots. This

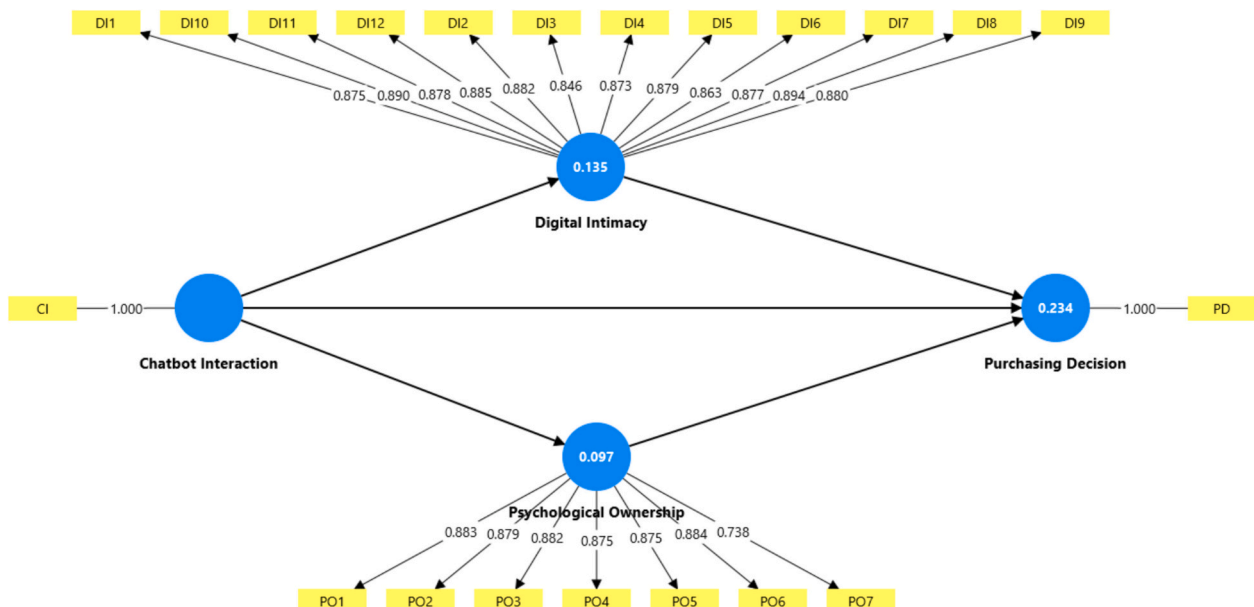


Fig. 2. Measurement model.

finding can be a cornerstone in developing technology-based interaction platforms that affect the psychological aspects of purchasing decisions. It can improve marketing and sales endeavors by offering interactions that will make customers happy and help them to feel closer to companies through technological tools.

An unexpected finding was the inability of AI chatbots to stimulate psychological ownership. There was a negative and significant impact of AI chatbot interaction on psychological ownership, confirming that humans can produce psychological ownership in customers much better than AI chatbots. This principle is connected to the relationship between customers and the products or services. Customers must be psychologically connected to the product to increase their desire for the product and encourage the purchasing decision. The chatbots' inability to create this feeling raises a question regarding their ability to imitate human interactions. This could be attributed to the inability of chatbots to create some important antecedents that contribute to the development of psychological ownership. Chatbots seem to fail at developing a sense of ownership because they don't make people feel they can touch the product or decide what happens to it. Moreover, Chatbots respond very quickly to customers; thus, this doesn't give customers time to feel they've invested their time or effort in that product. Furthermore, humans (salespeople) can usually create emotional history in their customers' minds. This seems not to be the case with the chatbots. Chatbots can't create the feeling of a long-term personal connection when interacting with customers.

Digital intimacy was found to have a significant effect on purchasing decisions. This shows the importance of considering humans' need to be treated and understood by a trusted partner. Customers felt closeness and a warm feeling in their interactions with AI chatbots, which makes it easier for them to complete their purchasing decision. This confirms the nature of the purchasing decision as a psychological action that depends on the person's feelings toward the company and the product itself.

Psychological ownership also had a significant effect on purchasing decisions. This clearly shows the gap in using AI chatbots as a customer interaction mechanism. Although psychological ownership is an important factor for purchasing decisions, AI chatbots failed to develop it. When customers feel that they are the owner of a product, failing to purchase that product is considered a loss, which makes the non-purchasing decision much harder. It is human nature for people to give more value to things they own, and thus, psychological ownership intensifies the feeling that the products or services have more value than the price paid for them.

The last finding of this study is the effect of AI chatbot interaction on the purchasing decision. The direct effect of AI chatbot interactions on purchasing decisions was found to be insignificant. This shows the importance of both digital intimacy and psychological ownership. The AI chatbot itself is not the key to encouraging purchasing decisions. Although new technologies are very useful for cost reduction and the creation of excitement among customers, this is not enough. Businesses should invest wisely in these technologies and prioritize customers' psychological behavior when designing such services. In particular, businesses should focus on (1) the creation of attractive media to interact with their customers and (2) persuading customers that the products or services are important for them. These two behaviors are represented in this study in the form of digital intimacy with the AI chatbots and psychological ownership of the product or service.

5.1. Theoretical implications

This study expanded existing knowledge on purchasing decisions and the use of new technology to alter these behaviors. By showing a fully mediating impact of digital intimacy and psychological ownership on the relationship between AI chatbot interactions and purchasing decisions, it revealed that the purchasing decision is a psychological behavior and presented new key factors that affect it. It also analyzed the interactions between humans and machines in the commercial context.

Previous studies have focused on the product and social norms as the main factors for a purchasing decision (Abdul Latip et al., 2025; Zhang et al., 2025). Using new technologies was presented as the goal of affecting the purchasing decision through technological advances and features (Cardinali et al., 2025; Palaniappan et al., 2025). This study shifted the attention to the psychological behavior of the customers.

This study advances this view by focusing on interactions with new technology and changes in customers' psychological behavior that can change the purchasing decision. AI chatbot technological characteristics were presented as the focal point of affecting customer behavior (Albayrak et al., 2018; Krishnan et al., 2022). Although this is a very insightful perspective, this study dug deeper into this direction and brought attention to the fact that these AI chatbot characteristics are only effective if they can affect the digital intimacy and psychological ownership of customers.

5.2. Practical implications

Small and medium-sized companies should give priority to the learning process of their AI chatbots. AI chatbots should be trained to embrace the psychological nature of customers and to make it fun for customers to interact with them, enhancing digital intimacy. AI chatbots should also be trained to enhance the customer's feeling toward the product or service by strengthening psychological ownership.

In addition, businesses should focus more on developing human-like AI chatbots (Zhang & Zhu, 2025) to secure more digital intimacy. The aim should be to create a customized experience for customers. Chatbots should put customers in usage scenarios that spark customers' imagination regarding how the product or service can change their lives. Training chatbots to answer customers' questions as an expert, thus creating the sense for the customer that he/she knows the product better than others, would also significantly enhance psychological ownership. Additionally, AI chatbots can provide real pictures of the products with more details, which can again improve psychological ownership for customers (Marckhgott et al., 2026).

6. Conclusion

This study aimed to understand the way that AI chatbot interactions affect customers' purchasing decisions. This was achieved through testing the dual mediating effects of customer digital intimacy and psychological ownership on this relationship. The study found fully dual mediation effects of digital intimacy and psychological ownership on the relationship between AI chatbot interactions and purchasing decisions.

6.1. Limitations

This study has some limitations. It adopted a cross-sectional design, which did not give the opportunity to see the effect of AI chatbot interactions on digital intimacy and psychological ownership over time. Additionally, this study did not differentiate digital intimacy from pseudo-digital intimacy because it assumed that humans consider AI chatbots to be feasible partners and are able to develop feelings while interacting with them. Studying pseudo-digital intimacy would focus on simulated closeness and emotional reactions in the context of loneliness. Finally, this study focused on individual customers instead of focusing on group or family customers. Studying family customers as a group can help to understand the mutual possession of product psychological ownership of more than one family/friend member. For example, strong psychological ownership of one member can have a negative effect on the level of psychological ownership of another member.

6.2. Future research

One of the main findings of this study is that AI chatbot interactions fail to develop psychological ownership. Future research is invited to dig

more in-depth into the reasons behind this failure. Moreover, the variables of this study, the AI chatbot interaction, digital intimacy, and psychological ownership, can interpret a relatively low variability in the purchasing decision. Thus, more moderating psychological variables can be studied in future research, such as social presence and self-congruity. Future research can adopt a longitudinal design to assess the relationship between AI chatbots, digital intimacy, and psychological ownership over time. Future research should also differentiate the psychological effects of digital intimacy and pseudo-digital intimacy. This would add value to our understanding of how AI chatbot interactions affect purchasing decisions in different types of customers. Studying the moderating effect of digital anxiety (Maduku et al., 2023) would also be useful. Intimacy, in general, is a psychological feeling that is usually connected to the biological needs of humans. Thus, digital intimacy can be considered human psychological manipulation that creates an illusory feeling. More studies are needed on the ethical considerations of creating this feeling in customers via an AI agent. More research is also needed to understand digital intimacy with new technologies and to discover the potential of humans to develop digital intimacy or pseudo-digital intimacy with AI chatbots. Finally, future research can focus on the interactions between AI chatbots, psychological ownership, and purchasing decisions for group customers instead of individual customers.

CRedit authorship contribution statement

Badr Alamr: Writing – review & editing, Writing – original draft, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization. **Abdulaziz Alharbi:** Writing – review & editing, Methodology, Investigation, Data curation. **Mohammed Ali Al-Awlaqi:** Writing – review & editing, Visualization, Software, Methodology.

Declaration of competing interest

The authors declare that they have no known financial or nonfinancial competing interests that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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